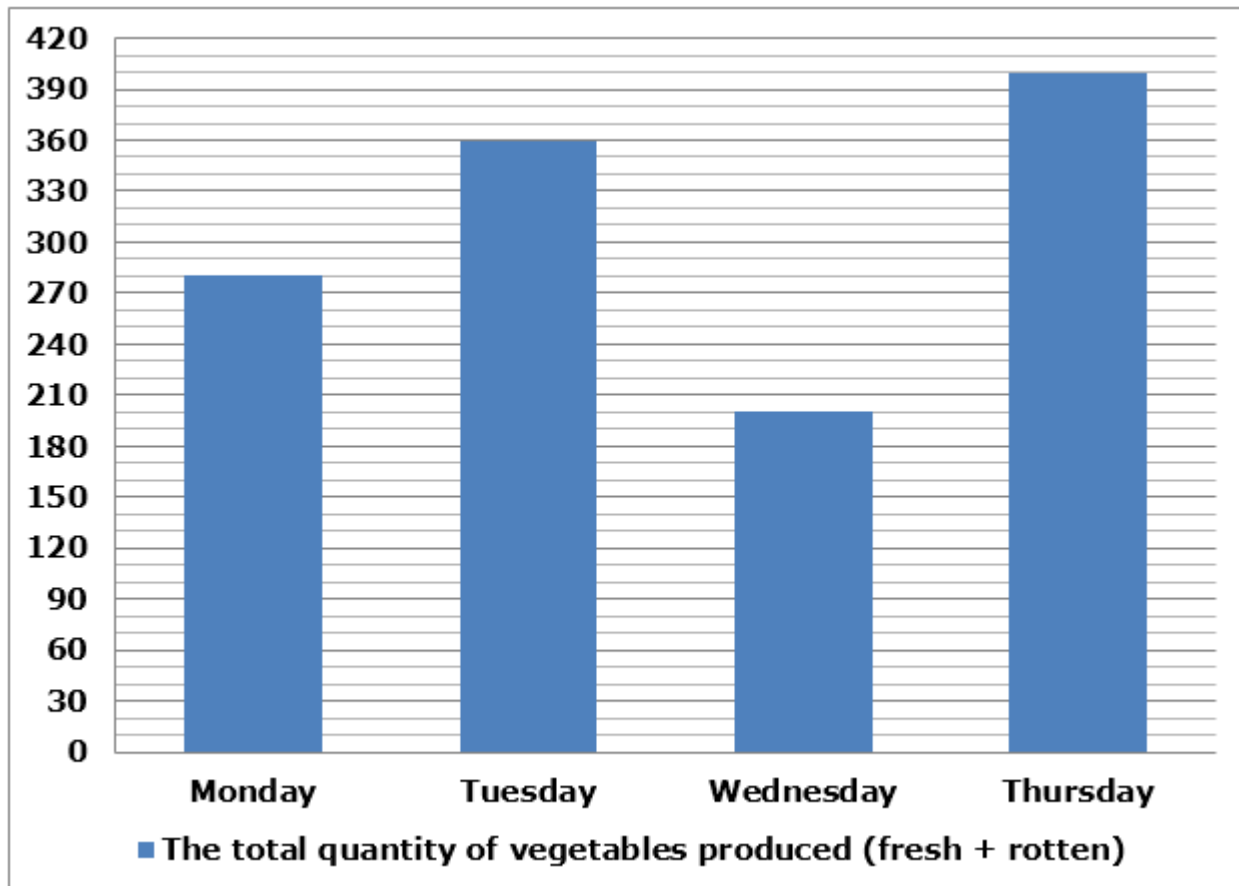


1. Questions

Read the following information carefully and answer the questions based on it.

The bar graph given below shows the total quantity (in kg) of vegetables produced (fresh + rotten) by a street vendor on four different days i.e. Monday, Tuesday, Wednesday and Thursday of a week. The table chart given below shows the percentage of the quantity of rotten vegetables out of the total quantity of vegetables produced and the quantity of fresh vegetables sold (in kg) on each day.



Days	The percentage of the quantity of rotten vegetables	The quantity of fresh vegetables sold (in kg)
Monday	$(X^2)\%$	
Tuesday		$(28Y+8)$
Wednesday	$3Y\%$	
Thursday		$(68X+2)$

Note:

Some data in the table chart are missing.

All the variables given in the table chart are positive integers.

The total quantity of vegetables produced = the quantity of fresh vegetables produced + the quantity of rotten vegetables produced

The total quantity of fresh vegetables produced = the quantity of fresh vegetables sold + the quantity of

fresh vegetables unsold

Only fresh vegetables can be sold.

The quantity of fresh vegetables produced on Thursday is $76X$ kg.

The ratio of the quantity of fresh vegetables produced on Monday is 190 kg more than the quantity of rotten vegetables produced on Thursday.

The quantity of fresh vegetables produced on Tuesday is $(30Y+24)$ kg.

The ratio of the quantity of rotten vegetables produced on Tuesday and fresh vegetables produced on Wednesday is 9:35.

On Monday, if the ratio of the quantity of fresh vegetables sold to males and females is 5:2, then the quantity of fresh vegetables unsold will be 30 kg less than the quantity of fresh vegetables sold to males. Find the total quantity of fresh vegetables sold (males + females) on Monday?

- a. 120 kg
- b. 140 kg
- c. 100 kg
- d. 160 kg
- e. None of these

2. Questions

On Tuesday, the ratio of the quantity of green and red fresh vegetables unsold is 5:4. If the quantity of red fresh vegetables sold is $\frac{M}{N}$ kg, which is 50% of the quantity of green fresh vegetables sold, then the difference between the quantity of green fresh vegetables sold and unsold will be $\frac{N}{M}$ kg. (Only green and red fresh vegetables are sold and unsold on Tuesday)

- a. $M < N$
- b. $M > N$
- c. $M \geq N$
- d. $M \leq N$
- e. $M = N$

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3. Questions

On Friday, if the total quantity of fresh vegetables produced is 120 kg, out of which 25% remained unsold. If the quantity of fresh vegetables which are sold on Friday is 3 times the quantity of rotten vegetables produced on the same day, then find the total quantity of vegetables (fresh + rotten) produced on Friday.

- a. 160 kg
- b. 240 kg

- c. 280 kg
- d. 150 kg
- e. None of these

4. Questions

On Thursday, if the total quantity of fruits produced (fresh + rotten) is [Q] and the quantity of rotten fruits produced is twice of the quantity of rotten vegetables produced on the same day, then the quantity of fresh fruits produced on Thursday will be [R] more than the quantity of rotten fruits produced on the same day. Find the total quantity of fruits produced (fresh + rotten) on Thursday. (The average of Q and R is 60)

- a. 100
- b. 50
- c. 120
- d. 80
- e. None of these

5. Questions

On Wednesday, the price per kg each vegetable is Rs. [T], and the total revenue generated from selling fresh vegetables is Rs. 1000. If the total quantity of fresh vegetables unsold on Wednesday is 20 less the quantity of rotten vegetables on the same day, then the value of [T] will be ____[?].

What should come in place of question mark?

- a. 12
- b. 24
- c. 30
- d. Cannot be determined
- e. None of these

6. Questions

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Read the following information carefully and answer the questions based on it.

Three shops P, Q and R sell two types of furniture i.e., tables and chairs.

Shop P: The ratio of the cost price of table and chair is 4:3. The table has been marked-up by 50% above its cost price and sold at a profit of Rs. 70. The selling price of the chair is $\frac{3}{5}$ th of the marked-price of the table, while the chair has been marked-up by Rs. 90. The discount allowed on the table is half of the discount allowed on the chair.

Shop Q: The selling price of the table is Rs. 280 more than the selling price of the chair. The cost price of the table is Rs. 400, which is sold at a discount of 11.11%. The marked price of the chair is Rs. 420, which

is sold at a profit of Rs. 60. The profit earned from selling the table is 4 times of the discount allowed on selling the chair.

Shop R: The marked price of the table is Rs. 20 more than the marked price of the chair. The discount allowed on selling table and chair is 16.66% and 25% respectively. The profit earned on selling table and chair is Rs. 70 and Rs. 60 respectively. The cost price of the chair is 14.28% less than the cost price of the table.

Note:

The profit should be calculated on the cost price, unless it is mentioned in the question.

In shop P, the total number of tables purchased by the shopkeeper is [4X], out of which 25% remained unsold. If the overall profit earned by the shopkeeper on selling all tables is Rs. 300, then find the total number of tables purchased by the shopkeeper from shop P.

- a. 160
- b. 180
- c. 120
- d. Cannot be determined
- e. None of these

7. Questions

In shop Q, if there is an additional discount of Rs. [D] is allowed, then the profit earned on selling the chair will be 6.25%, if calculated on selling price instead of the cost price. The value of [D] is ____[?]____. (The cost price and marked price remained unchanged)

What should come in place of question mark?

- a. Positive integer
- b. Exactly divisible by 10
- c. Composite number
- d. All of the above
- e. None of these

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8. Questions

The cost price of the table in another shop S is Rs. [Y], which is marked-up by 40% above its cost price and sold at a profit of Rs. 50 after allowing a discount of Rs. 10. Find the cost price of the chair in shop S, if the selling of the chair in the same shop is Rs. [Y+10], which is sold at a profit of 60%.

- a. Rs. 125
- b. Rs. 100
- c. Rs. 150

- d. Rs. 175
- e. None of these

9. Questions

In shop R, if the cost price of table and chair gets interchanged, while the rest of the things remained unchanged, then which of the following is/are definitely true?

- I:** The profit earned on selling the chair will be Rs. 15.
- II:** The mark-up percentage of the table will be 75.
- III:** The overall profit earned on selling table and chair will be 25%.

- a. Only I
- b. Both II and III
- c. Both I and II
- d. All I, II and III
- e. None is true

10. Questions

In shop Q, if the cost price of the table is increased by $[2Z]\%$ and marked-up by Rs. 60 above its cost price, then there will be a profit of 9.09%. If the discount allowed on the table is Rs. 20, then find the value of $[Z]$.

- a. 5
- b. 10
- c. 15
- d. 20
- e. None of these

11. Questions

Read the following information carefully and answer the questions based on it.

The table chart given below shows the number of people illiterate people in five different villages A, B, C, D and E. It also shows some information in the form of statement.

Village	The number of illiterate people	Statement
A	40	The number of illiterate people is $\frac{1}{3}$ rd of the number of post-graduate people
B	200	The number of graduate people is equal to the number of illiterate people
C	50	The number of post-graduate people is 20% more than the number of illiterate people
D	80	The ratio of the number of illiterate people and the number of post-graduate people is 2:3
E	32	The number of graduate people is 16 more than the number of post-graduate people

Note:

The total number of people in the village = the number of literate people + the number of illiterate people

The total number of literate people = the number of graduate people + the number of post-graduate people

Only literate people can be graduates and post-graduates.

The total number of people in village A is 25% less than village D.

The difference between number of graduate people in village A and D is 80. ($A < D$)

The ratio of the number of literate people in village B and C is 2:1.

The number of graduate people in village C is 10 less than the number of post-graduate people in village B.

The total number of people in village E is 160.

If the number of literate males in village B is _____, which is 80% of the number of literate males in village A, then the ratio of the number of literate females in village A and B will be _____.

Which of the following does satisfy the fillers in the same order?

I: 160 and 6:7

II: 144 and 8:9

III: 192 and 2:3

- Both I and II follow
- Only III follow
- Only I follow
- All I, II and III follow
- None follow

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12. Questions

In village C, out of the total number of graduate people, the number of people who are employed is $[X]$, while the rest of the graduates are unemployed. If the number of graduates who are employed in public sector is 3 times the number of unemployed graduates, while the number of employed graduates who are employed in private sector is 50, then find the value of $[\sqrt{X+1}]$. (Employed graduates are employed in private and public sector only)

- a. 3
- b. 9
- c. 12
- d. 16
- e. None of these

13. Questions

In another village F, the ratio of the number of literate and illiterate people is 5:1. If the number of graduates in village F is 120, which is $[Y]$, more than the number of post-graduates, then the total number of people in village F will be $[6Y]$.

Which of the following is/are definitely true about $[Y]$?

I: Exactly divisible by both 4 and 10.

II: Even composite number.

III: Unit digit is a prime number.

- a. Both I and II
- b. Both II and III
- c. Only III
- d. All I, II and III
- e. None follow

14. Questions

In village D, the ratio of the total number of people who are above and below poverty line is 3:1. If the number of females who are above poverty line is equal to the number of males who are below poverty line and ratio of the number of males who are above poverty line and females who are below poverty line is 7:1, then find the difference between the number of males and females who are above poverty line in village D.

- a. 120
- b. 200
- c. 180

- d. 240
- e. None of these

15. Questions

In village E, if the number of arts graduates is $\frac{1}{3}^{\text{rd}}$ of the number of commerce graduates, while the number of science graduates is 16 more than the number of commerce graduates, then find the ratio of the number of science and arts graduates in village E.

- a. 3:2
- b. 4:3
- c. 5:1
- d. Cannot be determined
- e. None of these

16. Questions

The following question has three statements as statement I, statement II and statement III. You have to determine whether data provided in three statements are sufficient to answer the question or not.

Find the difference between the initial quantity of water in vessel A and B.

Statement I: The ratio of the initial quantity of milk in vessel A and B is 5:4.

Statement II: The initial mixture (milk + water) in vessel B is 180 ml, which is 25% less than vessel A.

Statement III: The initial quantity of water in vessel A is 50% more than vessel B.

- a. Only statement I alone is sufficient
- b. Both statements I and II together are necessary
- c. Both statements I and III together are necessary
- d. All statements I, II and III together are necessary
- e. Neither statement I or II nor III is sufficient

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17. Questions

If the total work is 80 units, then find the time taken by R to complete the whole work alone.

Statement I: P and Q together can complete the whole work in $(80/9)$ days.

Statement II: Q and R together can complete the 70% of the total work in 8 days.

Statement III: The ratio of the efficiency of P and R is 2:1.

- a. Only statement I alone is sufficient
- b. Both statements I and II together are necessary

- c. Both statements I and III together are necessary
- d. All statements I, II and III together are necessary
- e. Neither statement I or II nor III is sufficient

18. Questions

Find the time taken by train Y to cross the platform of length 150 meters, if the speed of train X is 90 m/sec, which is 12.50% more than train Y.

Statement I: The ratio of the length of train X and Y is 6:5.

Statement II: The time taken by train X to cross a tunnel of length 60 meters in 4 seconds.

Statement III: Train X can cross a man running with the speed of 10 m/sec in opposite direction in 3 seconds.

- a. Only statement I alone is sufficient
- b. Both statements I and II together are necessary
- c. Either statement II or III together with statement I is sufficient
- d. All statements I, II and III together are necessary
- e. Neither statement I or II nor III is sufficient

19. Questions

Find the value of $\sqrt{R+5}$, if the ratio of the amount invested in scheme M and N is 4:5 and the rate of interest offered in scheme M is $[R]\%$ per annum compounded annually. (R is a positive integer)

Statement I: Scheme N offers simple interest at the rate of 15% per annum.

Statement II: The compound interest received from scheme M after 2 years is Rs. 17600.

Statement III: The simple interest received from scheme N after 3 years is Rs. 22500.

- a. Only statement I alone is sufficient
- b. Both statements I and II together are necessary
- c. Both statements I and III together are necessary
- d. All statements I, II and III together are necessary
- e. Neither statement I or II nor III is sufficient

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20. Questions

If boat A and B are travelling in the same river, whose speed is 3 km/hr, then find the speed of boat A in still water.

Statement I: The upstream speed of boat A is 80% of the downstream speed of boat B.

Statement II: The upstream speed of boat B is half of the downstream speed of boat A.

Statement III: The upstream distance covered by boat A in 5 hours is equal to the downstream distance covered by boat B in 4 hours.

- a. Only statement I alone is sufficient
- b. Both statements I and II together are necessary
- c. Either statement I and II together or II and III together is sufficient
- d. All statements I, II and III together are necessary
- e. Neither statement I or II nor III is sufficient

Explanations:

1. Questions

Monday

The total quantity of vegetables produced = 280 kg

The quantity of rotten vegetables produced = $(X^2)\%$ of 280 = $2.80X^2$ kg

The quantity of fresh vegetables produced = $(280 - 2.80X^2)$ kg

Thursday

The total quantity of vegetables produced = 400 kg

The quantity of fresh vegetables produced = $76X$ kg

The quantity of rotten vegetables produced = $(400 - 76X)$ kg

Now,

$$(280 - 2.80X^2) - (400 - 76X) = 190$$

$$76X - 120 - 2.80X^2 = 190$$

$$2.80X^2 - 76X + 310 = 0$$

$$28X^2 - 760X + 3100 = 0$$

$$7X^2 - 190X + 775 = 0$$

$$7X^2 - 155X - 35X + 775 = 0$$

$$7X(X - 22.14) - 35(X - 22.14) = 0$$

$$(7X - 35)(X - 22.14) = 0$$

$$X = +5 \text{ or } +22.14$$

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So, $X = 5$ (positive integer)

Tuesday

The total quantity of vegetables produced = 360 kg

The quantity of fresh vegetables produced = $(30Y+24)$ kg

The quantity of rotten vegetables produced = $\{360-(30Y+24)\} = (336-30Y)$ kg

Wednesday

The total quantity of vegetables produced = 200 kg

The quantity of rotten vegetables produced = $3Y\%$ of 200 = $6Y$ kg

The quantity of fresh vegetables produced = $(200-6Y)$ kg

Now,

$$(336-30Y)/(200-6Y) = 9/35$$

$$11760 - 1050Y = 1800 - 54Y$$

$$996Y = 9960$$

$$Y = 10$$

Monday

The quantity of rotten vegetables produced = $2.80X^2 = (2.80 \times 5^2) = (2.80 \times 25) = 70$ kg

The quantity of fresh vegetables produced = $(280-2.80X^2) = (280-70) = 210$ kg

Tuesday

The quantity of fresh vegetables produced = $(30Y+24) = (30 \times 10+24) = (300+24) = 324$ kg

The quantity of rotten vegetables produced = $(336-30Y) = (336-300) = 36$ kg

The quantity of fresh vegetables sold = $(28Y+8) = (28 \times 10+8) = (280+8) = 288$ kg

The quantity of fresh vegetables unsold = $(324-288) = 36$ kg

Wednesday

The quantity of rotten vegetables produced = $6Y$ kg = $(6 \times 10) = 60$ kg

The quantity of fresh vegetables produced = $(200-6Y) = (200-60) = 140$ kg

Thursday

The quantity of fresh vegetables produced = $76X = (76 \times 5) = 380$ kg

The quantity of rotten vegetables produced = $(400-76X) = (400-380) = 20$ kg

The quantity of fresh vegetables sold = $(68 \times 5+2) = (340+2) = 342$ kg

The quantity of fresh vegetables unsold = $(380-342) = 38$ kg

Answer: B

Now, according to the question,

Monday

The total quantity of fresh vegetables produced = 210 kg

Now, let the quantity of fresh vegetables sold to males and females be $5P$ kg and $2P$ kg respectively.

So, the total quantity of fresh vegetables sold = $7P$ kg

So, the total quantity of fresh vegetables unsold = $(210 - 7P)$ kg

The quantity of fresh vegetables sold to males = $\frac{5}{7}$ of $7P = 5P$ kg

The quantity of fresh vegetables sold to females = $\frac{2}{7}$ of $7P = 2P$ kg

Now,

$$5P - (210 - 7P) = 30$$

$$5P - 30 = 210 - 7P$$

$$12P = 240$$

$$P = 20$$

Therefore, the total quantity of fresh vegetables sold = $7P = (7 \times 20) = 140$ kg

Hence, the correct answer is option B.

2. Questions

Monday

The total quantity of vegetables produced = 280 kg

The quantity of rotten vegetables produced = $(X^2)\%$ of 280 = $2.80X^2$ kg

The quantity of fresh vegetables produced = $(280 - 2.80X^2)$ kg

Thursday

The total quantity of vegetables produced = 400 kg

The quantity of fresh vegetables produced = $76X$ kg

The quantity of rotten vegetables produced = $(400 - 76X)$ kg

Now,

$$(280 - 2.80X^2) - (400 - 76X) = 190$$

$$76X - 120 - 2.80X^2 = 190$$

$$2.80X^2 - 76X + 310 = 0$$

$$28X^2 - 760X + 3100 = 0$$

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$$7X^2 - 190X + 775 = 0$$

$$7X^2 - 155X - 35X + 775 = 0$$

$$7X(X-22.14) - 35(X-22.14) = 0$$

$$(7X-35)(X-22.14) = 0$$

$$X = +5 \text{ or } +22.14$$

So, $X = 5$ (positive integer)

Tuesday

The total quantity of vegetables produced = 360 kg

The quantity of fresh vegetables produced = $(30Y+24)$ kg

The quantity of rotten vegetables produced = $\{360-(30Y+24)\} = (336-30Y)$ kg

Wednesday

The total quantity of vegetables produced = 200 kg

The quantity of rotten vegetables produced = $3Y\%$ of 200 = $6Y$ kg

The quantity of fresh vegetables produced = $(200-6Y)$ kg

Now,

$$(336-30Y)/(200-6Y) = 9/35$$

$$11760 - 1050Y = 1800 - 54Y$$

$$996Y = 9960$$

$$Y = 10$$

Monday

The quantity of rotten vegetables produced = $2.80X^2 = (2.80 \times 5^2) = (2.80 \times 25) = 70$ kg

The quantity of fresh vegetables produced = $(280-2.80X^2) = (280-70) = 210$ kg

Tuesday

The quantity of fresh vegetables produced = $(30Y+24) = (30 \times 10 + 24) = (300+24) = 324$ kg

The quantity of rotten vegetables produced = $(336-30Y) = (336-300) = 36$ kg

The quantity of fresh vegetables sold = $(28Y+8) = (28 \times 10 + 8) = (280+8) = 288$ kg

The quantity of fresh vegetables unsold = $(324-288) = 36$ kg

Wednesday

The quantity of rotten vegetables produced = $6Y$ kg = $(6 \times 10) = 60$ kg

The quantity of fresh vegetables produced = $(200-6Y) = (200-60) = 140$ kg

Thursday

The quantity of fresh vegetables produced = $76X = (76 \times 5) = 380$ kg

The quantity of rotten vegetables produced = $(400 - 76X) = (400 - 380) = 20$ kg

The quantity of fresh vegetables sold = $(68 \times 5 + 2) = (340 + 2) = 342$ kg

The quantity of fresh vegetables unsold = $(380 - 342) = 38$ kg

Answer: A

Tuesday

The total quantity of fresh vegetables (green + red) unsold = 36 kg

The quantity of green fresh vegetables unsold = $\frac{5}{9}$ of 36 = 20 kg

The quantity of red fresh vegetables unsold = $\frac{4}{9}$ of 36 = 16 kg

The total quantity of fresh vegetables (green + red) sold = 288 kg

Now, the ratio of the quantity of green and red fresh vegetables sold = $100:50 = 2:1$

The quantity of green fresh vegetables sold = $\frac{2}{3}$ of 288 = 192 kg

The quantity of red fresh vegetables sold = $\frac{1}{3}$ of 288 = 96 kg

So, $M = 96$

Therefore, the difference between quantity of green fresh vegetables sold and unsold = $(192 - 20) = 172$ kg

So, $N = 172$

Therefore, $M < N$.

Hence, the correct answer is option A.

3. Questions

Monday

The total quantity of vegetables produced = 280 kg

The quantity of rotten vegetables produced = $(X^2)\%$ of 280 = $2.80X^2$ kg

The quantity of fresh vegetables produced = $(280 - 2.80X^2)$ kg

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Thursday

The total quantity of vegetables produced = 400 kg

The quantity of fresh vegetables produced = $76X$ kg

The quantity of rotten vegetables produced = $(400 - 76X)$ kg

Now,

$$(280 - 2.80X^2) - (400 - 76X) = 190$$

$$76X - 120 - 2.80X^2 = 190$$

$$2.80X^2 - 76X + 310 = 0$$

$$28X^2 - 760X + 3100 = 0$$

$$7X^2 - 190X + 775 = 0$$

$$7X^2 - 155X - 35X + 775 = 0$$

$$7X(X-22.14) - 35(X-22.14) = 0$$

$$(7X-35)(X-22.14) = 0$$

$$X = +5 \text{ or } +22.14$$

So, $X = 5$ (positive integer)

Tuesday

The total quantity of vegetables produced = 360 kg

The quantity of fresh vegetables produced = $(30Y+24)$ kg

The quantity of rotten vegetables produced = $\{360-(30Y+24)\} = (336-30Y)$ kg

Wednesday

The total quantity of vegetables produced = 200 kg

The quantity of rotten vegetables produced = $3Y\%$ of 200 = $6Y$ kg

The quantity of fresh vegetables produced = $(200-6Y)$ kg

Now,

$$(336-30Y)/(200-6Y) = 9/35$$

$$11760 - 1050Y = 1800 - 54Y$$

$$996Y = 9960$$

$$Y = 10$$

Monday

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The quantity of rotten vegetables produced = $2.80X^2 = (2.80 \times 5^2) = (2.80 \times 25) = 70$ kg

The quantity of fresh vegetables produced = $(280-2.80X^2) = (280-70) = 210$ kg

Tuesday

The quantity of fresh vegetables produced = $(30Y+24) = (30 \times 10+24) = (300+24) = 324$ kg

The quantity of rotten vegetables produced = $(336-30Y) = (336-300) = 36$ kg

The quantity of fresh vegetables sold = $(28Y+8) = (28 \times 10+8) = (280+8) = 288$ kg

The quantity of fresh vegetables unsold = $(324-288) = 36$ kg

Wednesday

The quantity of rotten vegetables produced = $6Y$ kg = $(6 \times 10) = 60$ kg

The quantity of fresh vegetables produced = $(200-6Y) = (200-60) = 140$ kg

Thursday

The quantity of fresh vegetables produced = $76X = (76 \times 5) = 380$ kg

The quantity of rotten vegetables produced = $(400-76X) = (400-380) = 20$ kg

The quantity of fresh vegetables sold = $(68 \times 5 + 2) = (340 + 2) = 342$ kg

The quantity of fresh vegetables unsold = $(380-342) = 38$ kg

Answer: D

Friday

The total quantity of fresh vegetables produced = 120 kg

The quantity of fresh vegetables unsold = 25% of 120 = 30 kg

The quantity of fresh vegetables sold = 75% of 120 = 90 kg

So, the quantity of rotten vegetables produced = $(90/3) = 30$ kg

Therefore, the total quantity of vegetables (fresh + rotten) produced on Friday = $(120+30) = 150$ kg.

Hence, the correct answer is option D.

4. Questions

Monday

The total quantity of vegetables produced = 280 kg

The quantity of rotten vegetables produced = $(X^2)\%$ of 280 = $2.80X^2$ kg

The quantity of fresh vegetables produced = $(280-2.80X^2)$ kg

Thursday

The total quantity of vegetables produced = 400 kg

The quantity of fresh vegetables produced = $76X$ kg

The quantity of rotten vegetables produced = $(400-76X)$ kg

Now,

$$(280-2.80X^2) - (400-76X) = 190$$

$$76X - 120 - 2.80X^2 = 190$$

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$$7X^2 - 155X - 35X + 775 = 0$$

$$7X(X-22.14) - 35(X-22.14) = 0$$

$$(7X-35)(X-22.14) = 0$$

$$X = +5 \text{ or } +22.14$$

So, $X = 5$ (positive integer)

Tuesday

The total quantity of vegetables produced = 360 kg

The quantity of fresh vegetables produced = $(30Y+24)$ kg

The quantity of rotten vegetables produced = $\{360-(30Y+24)\} = (336-30Y)$ kg

Wednesday

The total quantity of vegetables produced = 200 kg

The quantity of rotten vegetables produced = $3Y\%$ of 200 = $6Y$ kg

The quantity of fresh vegetables produced = $(200-6Y)$ kg

Now,

$$(336-30Y)/(200-6Y) = 9/35$$

$$11760 - 1050Y = 1800 - 54Y$$

$$996Y = 9960$$

$$Y = 10$$

Monday

The quantity of rotten vegetables produced = $2.80X^2 = (2.80 \times 5^2) = (2.80 \times 25) = 70$ kg

The quantity of fresh vegetables produced = $(280-2.80X^2) = (280-70) = 210$ kg

Tuesday

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Wednesday

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Thursday

The quantity of fresh vegetables produced = $76X = (76 \times 5) = 380$ kg

The quantity of rotten vegetables produced = $(400 - 76X) = (400 - 380) = 20$ kg

The quantity of fresh vegetables sold = $(68 \times 5 + 2) = (340 + 2) = 342$ kg

The quantity of fresh vegetables unsold = $(380 - 342) = 38$ kg

Answer: A

Thursday

The quantity of rotten vegetables produced = 20

The total quantity of fruits produced = Q

The quantity of rotten fruits produced = $(20 \times 2) = 40$

So, the quantity of fresh fruits produced = $(Q - 40)$

Now,

$$(Q - 40) - 40 = R$$

$$Q - 80 = R$$

$$Q - R = 80 \dots\dots\dots(i)$$

Also,

$$Q + R = (60 \times 2)$$

$$Q + R = 120 \dots\dots\dots(ii)$$

By solving equations (i) and (ii), we get,

$$2R = 40$$

$$R = 20$$

Now, put the value of R in equation (i), we get,

$$Q - R = 80$$

$$Q - 20 = 80$$

$$Q = 100$$

Therefore, total quantity of fruits produced on Thursday = $Q = 100$.

Hence, the correct answer is option A.

5. Questions

Monday

The total quantity of vegetables produced = 280 kg

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The quantity of rotten vegetables produced = $(X^2)\%$ of 280 = $2.80X^2$ kg

The quantity of fresh vegetables produced = $(280 - 2.80X^2)$ kg

Thursday

The total quantity of vegetables produced = 400 kg

The quantity of fresh vegetables produced = $76X$ kg

The quantity of rotten vegetables produced = $(400 - 76X)$ kg

Now,

$$(280 - 2.80X^2) - (400 - 76X) = 190$$

$$76X - 120 - 2.80X^2 = 190$$

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$$28X^2 - 760X + 3100 = 0$$

$$7X^2 - 190X + 775 = 0$$

$$7X^2 - 155X - 35X + 775 = 0$$

$$7X(X - 22.14) - 35(X - 22.14) = 0$$

$$(7X - 35)(X - 22.14) = 0$$

$$X = +5 \text{ or } +22.14$$

So, $X = 5$ (positive integer)

Tuesday

The total quantity of vegetables produced = 360 kg

The quantity of fresh vegetables produced = $(30Y + 24)$ kg

The quantity of rotten vegetables produced = $\{360 - (30Y + 24)\} = (336 - 30Y)$ kg

Wednesday

The total quantity of vegetables produced = 200 kg

The quantity of rotten vegetables produced = $3Y\%$ of 200 = $6Y$ kg

The quantity of fresh vegetables produced = $(200 - 6Y)$ kg

Now,

$$(336 - 30Y) / (200 - 6Y) = 9/35$$

$$11760 - 1050Y = 1800 - 54Y$$

$$996Y = 9960$$

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$$Y = 10$$

Monday

The quantity of rotten vegetables produced = $2.80X^2 = (2.80 \times 5^2) = (2.80 \times 25) = 70$ kg

The quantity of fresh vegetables produced = $(280 - 2.80X^2) = (280 - 70) = 210$ kg

Tuesday

The quantity of fresh vegetables produced = $(30Y + 24) = (30 \times 10 + 24) = (300 + 24) = 324$ kg

The quantity of rotten vegetables produced = $(336 - 30Y) = (336 - 300) = 36$ kg

The quantity of fresh vegetables sold = $(28Y + 8) = (28 \times 10 + 8) = (280 + 8) = 288$ kg

The quantity of fresh vegetables unsold = $(324 - 288) = 36$ kg

Wednesday

The quantity of rotten vegetables produced = $6Y$ kg = $(6 \times 10) = 60$ kg

The quantity of fresh vegetables produced = $(200 - 6Y) = (200 - 60) = 140$ kg

Thursday

The quantity of fresh vegetables produced = $76X = (76 \times 5) = 380$ kg

The quantity of rotten vegetables produced = $(400 - 76X) = (400 - 380) = 20$ kg

The quantity of fresh vegetables sold = $(68 \times 5 + 2) = (340 + 2) = 342$ kg

The quantity of fresh vegetables unsold = $(380 - 342) = 38$ kg

Answer: E

Wednesday

The total quantity of vegetables produced = 200 kg

The total quantity of fresh vegetables produced = 140 kg

The quantity of rotten vegetables produced = 60

So, the quantity of fresh vegetables unsold = $(60 - 20) = 40$ kg

Now, the quantity of fresh vegetables sold = $(140 - 40) = 100$ kg @QuizByMickey

And, the total revenue generated from selling fresh vegetables = Rs. 1000

Therefore, the price per kg each vegetable = $(1000/100) = \text{Rs. } 10$

So, the value of T = 10.

Hence, the correct answer is option E.

6. Questions

Shop P

Let the C.P. of table and chair be Rs. 4m and Rs. 3m respectively.

The C.P. of the table = Rs. $4m$

The M.P. of the table = 150% of $4m$ = Rs. $6m$

The S.P. of the table = Rs. $(4m+70)$

The discount allowed on the table = $\{6m-(4m+70)\}$ = Rs. $(2m-70)$

The C.P. of the chair = Rs. $3m$

The M.P. of the chair = Rs. $(3m+90)$

The S.P. of the chair = $\frac{3}{5}$ of $6m$ = Rs. $3.60m$

The discount allowed on the chair = $\{(3m+90)-3.60m\}$ = Rs. $(90-0.60m)$

Now,

$$(2m-70)/(90-0.60m) = 1/2$$

$$4m - 140 = 90 - 0.60m$$

$$4.60m = 230$$

$$m = 50$$

Therefore,

The C.P. of the table = $4m = (4 \times 50)$ = Rs. 200

The M.P. of the table = $6m = (6 \times 50)$ = Rs. 300

The S.P. of the table = $(4m+70) = (200+70)$ = Rs. 270

The C.P. of the chair = $3m = (3 \times 50)$ = Rs. 150

The M.P. of the chair = $(3m+90) = (150+90)$ = Rs. 240

The S.P. of the chair = $3.60m = (3.60 \times 50)$ = Rs. 180

Shop Q

Let the M.P. of the table be Rs. $9n$. (*Note: 11.11% = 1/9*)

The C.P. of the table = Rs. 400

The M.P. of the table = Rs. $9n$

The S.P. of the table = $\{(9n/9) \times 8\}$ = Rs. $8n$

The profit earned on selling the table = Rs. $(8n-400)$

The S.P. of the chair = Rs. $(8n-280)$

The C.P. of the chair = $\{(8n-280)-60\}$ = Rs. $(8n-340)$

The M.P. of the chair = Rs. 420

The discount allowed on selling the chair = $\{420-(8n-280)\}$ = Rs. $(700-8n)$

Now,

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$$(8n-400)/(700-8n) = 4/1$$

$$8n - 400 = 2800 - 32n$$

$$40n = 3200$$

$$n = 80$$

Therefore,

$$\text{The C.P. of the table} = 400$$

$$\text{The M.P. of the table} = 9n = (9 \times 80) = 720$$

$$\text{The S.P. of the table} = 8n = (8 \times 80) = \text{Rs. } 640$$

$$\text{The S.P. of the chair} = (8n-280) = (640-280) = \text{Rs. } 360$$

$$\text{The C.P. of the chair} = (8n-340) = (640-340) = \text{Rs. } 300$$

Shop R

Let the marked price of the table be Rs. 6k (*Note: 16.66% = 1/6*)

$$\text{The S.P. of the table} = \{(6k/6) \times 5\} = \text{Rs. } 5k$$

$$\text{The C.P. of the table} = \text{Rs. } (5k-70)$$

$$\text{The M.P. of the chair} = \text{Rs. } (6k-20)$$

$$\text{The S.P. of the chair} = 75\% \text{ of } (6k-20) = \text{Rs. } (4.50k-15)$$

$$\text{The C.P. of the chair} = \{(4.50k-15)-60\} = \text{Rs. } (4.50k-75)$$

Now, the ratio of the cost price of table and chair = 7:6 (*Note: 14.28% = 1/7*)

$$(5k-70)/(4.50k-75) = 7/6$$

$$30k - 420 = 31.50k - 525$$

$$1.50k = 105$$

$$k = 70$$

Therefore,

$$\text{The M.P. of the table} = 6k = (6 \times 70) = \text{Rs. } 420$$

$$\text{The S.P. of the table} = 5k = (5 \times 70) = \text{Rs. } 350$$

$$\text{The C.P. of the table} = (5k-70) = (350-70) = \text{Rs. } 280$$

$$\text{The M.P. of the chair} = (6k-20) = (420-20) = \text{Rs. } 400$$

$$\text{The S.P. of the chair} = (4.50k-15) = (4.50 \times 70 - 15) = (315-15) = \text{Rs. } 300$$

$$\text{The C.P. of the chair} = (4.50k-75) = (315-75) = \text{Rs. } 240$$

Shop P

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Furniture	The cost price (in Rs.)	The marked price (in Rs.)	The selling price (in Rs.)
Table	200	300	270
Chair	150	240	180
Total	350	540	450

Shop Q

Furniture	The cost price (in Rs.)	The marked price (in Rs.)	The selling price (in Rs.)
Table	400	720	640
Chair	300	420	360
Total	700	1140	1000

Shop R

Furniture	The cost price (in Rs.)	The marked price (in Rs.)	The selling price (in Rs.)
Table	280	420	350
Chair	240	400	300
Total	520	820	650

Answer: C

Now, according to the question,

Shop P

The total number of tables purchased = $4X$

The number of tables unsold = 25% of $4X = X$

So, the total number of tables sold = $(4X - X) = 3X$

The C.P. of each table = Rs. 200

The S.P. of each table = Rs. 270

So, the overall C.P. of all tables purchased = $(4X \times 200) = \text{Rs. } 800X$

And, the overall S.P. of all tables sold = $(3X \times 270) = \text{Rs. } 810X$

Now,

$$810X - 800X = 300$$

$$10X = 300$$

$$X = 30$$

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Therefore, total number of tables purchased by the shopkeeper from shop P = $4X = (4 \times 30) = 120$

Hence, the correct answer is option C.

7. Questions

Shop P

Let the C.P. of table and chair be Rs. $4m$ and Rs. $3m$ respectively.

The C.P. of the table = Rs. $4m$

The M.P. of the table = 150% of $4m$ = Rs. $6m$

The S.P. of the table = Rs. $(4m+70)$

The discount allowed on the table = $\{6m-(4m+70)\}$ = Rs. $(2m-70)$

The C.P. of the chair = Rs. $3m$

The M.P. of the chair = Rs. $(3m+90)$

The S.P. of the chair = $\frac{3}{5}$ of $6m$ = Rs. $3.60m$

The discount allowed on the chair = $\{(3m+90)-3.60m\}$ = Rs. $(90-0.60m)$

Now,

$$(2m-70)/(90-0.60m) = 1/2$$

$$4m - 140 = 90 - 0.60m$$

$$4.60m = 230$$

$$m = 50$$

Therefore,

The C.P. of the table = $4m = (4 \times 50)$ = Rs. 200

The M.P. of the table = $6m = (6 \times 50)$ = Rs. 300

The S.P. of the table = $(4m+70) = (200+70)$ = Rs. 270

The C.P. of the chair = $3m = (3 \times 50)$ = Rs. 150

The M.P. of the chair = $(3m+90) = (150+90)$ = Rs. 240

The S.P. of the chair = $3.60m = (3.60 \times 50)$ = Rs. 180

Shop Q

Let the M.P. of the table be Rs. $9n$. (**Note: $11.11\% = 1/9$**)

The C.P. of the table = Rs. 400

The M.P. of the table = Rs. $9n$

The S.P. of the table = $\{(9n/9) \times 8\}$ = Rs. $8n$

The profit earned on selling the table = Rs. $(8n-400)$

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The S.P. of the chair = Rs. $(8n-280)$

The C.P. of the chair = $\{(8n-280)-60\}$ = Rs. $(8n-340)$

The M.P. of the chair = Rs. 420

The discount allowed on selling the chair = $\{420-(8n-280)\}$ = Rs. $(700-8n)$

Now,

$$(8n-400)/(700-8n) = 4/1$$

$$8n - 400 = 2800 - 32n$$

$$40n = 3200$$

$$n = 80$$

Therefore,

The C.P. of the table = 400

The M.P. of the table = $9n = (9 \times 80) = 720$

The S.P. of the table = $8n = (8 \times 80) = \text{Rs. } 640$

The S.P. of the chair = $(8n-280) = (640-280) = \text{Rs. } 360$

The C.P. of the chair = $(8n-340) = (640-340) = \text{Rs. } 300$

Shop R

Let the marked price of the table be Rs. $6k$ (*Note: $16.66\% = 1/6$*)

The S.P. of the table = $\{(6k/6) \times 5\} = \text{Rs. } 5k$

The C.P. of the table = Rs. $(5k-70)$

The M.P. of the chair = Rs. $(6k-20)$

The S.P. of the chair = 75% of $(6k-20) = \text{Rs. } (4.50k-15)$

The C.P. of the chair = $\{(4.50k-15)-60\} = \text{Rs. } (4.50k-75)$

Now, the ratio of the cost price of table and chair = $7:6$ (*Note: $14.28\% = 1/7$*)

$$(5k-70)/(4.50k-75) = 7/6$$

$$30k - 420 = 31.50k - 525$$

$$1.50k = 105$$

$$k = 70$$

Therefore,

The M.P. of the table = $6k = (6 \times 70) = \text{Rs. } 420$

The S.P. of the table = $5k = (5 \times 70) = \text{Rs. } 350$

The C.P. of the table = $(5k-70) = (350-70) = \text{Rs. } 280$

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The M.P. of the chair = $(6k-20) = (420-20) = \text{Rs. } 400$

The S.P. of the chair = $(4.50k-15) = (4.50 \times 70-15) = (315-15) = \text{Rs. } 300$

The C.P. of the chair = $(4.50k-75) = (315-75) = \text{Rs. } 240$

Shop P

Furniture	The cost price (in Rs.)	The marked price (in Rs.)	The selling price (in Rs.)
Table	200	300	270
Chair	150	240	180
Total	350	540	450

Shop Q

Furniture	The cost price (in Rs.)	The marked price (in Rs.)	The selling price (in Rs.)
Table	400	720	640
Chair	300	420	360
Total	700	1140	1000

Shop R

Furniture	The cost price (in Rs.)	The marked price (in Rs.)	The selling price (in Rs.)
Table	280	420	350
Chair	240	400	300
Total	520	820	650

Answer: D

Shop Q (Chair)

The C.P. = Rs. 300

The S.P. after additional discount = Rs. $(360-D)$

The fractional value of 6.25 = $1/16$

So, the ratio of the C.P. and the S.P. = 15:16

Now,

$$300/(360-D) = 15/16$$

$$4800 = 5400 - 15D$$

$$15D = 600$$

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D = 40

40 is a positive composite integer, which is exactly divisible by 10 and also a composite number ($5 \times 8 = 40$).

Hence, the correct answer is option D.

8. Questions

Shop P

Let the C.P. of table and chair be Rs. $4m$ and Rs. $3m$ respectively.

The C.P. of the table = Rs. $4m$

The M.P. of the table = 150% of $4m$ = Rs. $6m$

The S.P. of the table = Rs. $(4m+70)$

The discount allowed on the table = $\{6m - (4m+70)\}$ = Rs. $(2m-70)$

The C.P. of the chair = Rs. $3m$

The M.P. of the chair = Rs. $(3m+90)$

The S.P. of the chair = $\frac{3}{5}$ of $6m$ = Rs. $3.60m$

The discount allowed on the chair = $\{(3m+90) - 3.60m\}$ = Rs. $(90-0.60m)$

Now,

$$(2m-70)/(90-0.60m) = 1/2$$

$$4m - 140 = 90 - 0.60m$$

$$4.60m = 230$$

$$m = 50$$

Therefore,

The C.P. of the table = $4m = (4 \times 50)$ = Rs. 200

The M.P. of the table = $6m = (6 \times 50)$ = Rs. 300

The S.P. of the table = $(4m+70) = (200+70)$ = Rs. 270

The C.P. of the chair = $3m = (3 \times 50)$ = Rs. 150

The M.P. of the chair = $(3m+90) = (150+90)$ = Rs. 240

The S.P. of the chair = $3.60m = (3.60 \times 50)$ = Rs. 180

Shop Q

Let the M.P. of the table be Rs. $9n$. (**Note: $11.11\% = 1/9$**)

The C.P. of the table = Rs. 400

The M.P. of the table = Rs. $9n$

The S.P. of the table = $\{(9n/9) \times 8\}$ = Rs. $8n$

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The profit earned on selling the table = Rs. $(8n-400)$

The S.P. of the chair = Rs. $(8n-280)$

The C.P. of the chair = $\{(8n-280)-60\}$ = Rs. $(8n-340)$

The M.P. of the chair = Rs. 420

The discount allowed on selling the chair = $\{420-(8n-280)\}$ = Rs. $(700-8n)$

Now,

$$(8n-400)/(700-8n) = 4/1$$

$$8n - 400 = 2800 - 32n$$

$$40n = 3200$$

$$n = 80$$

Therefore,

The C.P. of the table = 400

The M.P. of the table = $9n = (9 \times 80) = 720$

The S.P. of the table = $8n = (8 \times 80) = \text{Rs. } 640$

The S.P. of the chair = $(8n-280) = (640-280) = \text{Rs. } 360$

The C.P. of the chair = $(8n-340) = (640-340) = \text{Rs. } 300$

Shop R

Let the marked price of the table be Rs. $6k$ (*Note: $16.66\% = 1/6$*)

The S.P. of the table = $\{(6k/6) \times 5\} = \text{Rs. } 5k$

The C.P. of the table = Rs. $(5k-70)$

The M.P. of the chair = Rs. $(6k-20)$

The S.P. of the chair = 75% of $(6k-20) = \text{Rs. } (4.50k-15)$

The C.P. of the chair = $\{(4.50k-15)-60\} = \text{Rs. } (4.50k-75)$

Now, the ratio of the cost price of table and chair = $7:6$ (*Note: $14.28\% = 1/7$*)

$$(5k-70)/(4.50k-75) = 7/6$$

$$30k - 420 = 31.50k - 525$$

$$1.50k = 105$$

$$k = 70$$

Therefore,

The M.P. of the table = $6k = (6 \times 70) = \text{Rs. } 420$

The S.P. of the table = $5k = (5 \times 70) = \text{Rs. } 350$

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The C.P. of the table = $(5k-70) = (350-70) = \text{Rs. } 280$

The M.P. of the chair = $(6k-20) = (420-20) = \text{Rs. } 400$

The S.P. of the chair = $(4.50k-15) = (4.50 \times 70 - 15) = (315-15) = \text{Rs. } 300$

The C.P. of the chair = $(4.50k-75) = (315-75) = \text{Rs. } 240$

Shop P

Furniture	The cost price (in Rs.)	The marked price (in Rs.)	The selling price (in Rs.)
Table	200	300	270
Chair	150	240	180
Total	350	540	450

Shop Q

Furniture	The cost price (in Rs.)	The marked price (in Rs.)	The selling price (in Rs.)
Table	400	720	640
Chair	300	420	360
Total	700	1140	1000

Shop R

Furniture	The cost price (in Rs.)	The marked price (in Rs.)	The selling price (in Rs.)
Table	280	420	350
Chair	240	400	300
Total	520	820	650

Answer: B

Shop S

Table

The C.P. = Rs. Y

The M.P. = 140% of Y = Rs. 1.40Y

The S.P. = Rs. (Y+50)

Now,

$$1.40Y - (Y+50) = 10$$

$$1.40Y - 10 = Y + 50$$

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$$0.40Y = 60$$

$$Y = 150$$

So, the S.P. of the chair = $(Y+10) = (150+10) = \text{Rs. } 160$

Therefore, the C.P. of the chair = $\{(160/160) \times 100\} = \text{Rs. } 100$

Hence, the correct answer is option B.

9. Questions

Shop P

Let the C.P. of table and chair be Rs. $4m$ and Rs. $3m$ respectively.

The C.P. of the table = Rs. $4m$

The M.P. of the table = 150% of $4m = \text{Rs. } 6m$

The S.P. of the table = Rs. $(4m+70)$

The discount allowed on the table = $\{6m - (4m+70)\} = \text{Rs. } (2m-70)$

The C.P. of the chair = Rs. $3m$

The M.P. of the chair = Rs. $(3m+90)$

The S.P. of the chair = $3/5$ of $6m = \text{Rs. } 3.60m$

The discount allowed on the chair = $\{(3m+90) - 3.60m\} = \text{Rs. } (90-0.60m)$

Now,

$$(2m-70)/(90-0.60m) = 1/2$$

$$4m - 140 = 90 - 0.60m$$

$$4.60m = 230$$

$$m = 50$$

Therefore,

The C.P. of the table = $4m = (4 \times 50) = \text{Rs. } 200$

The M.P. of the table = $6m = (6 \times 50) = \text{Rs. } 300$

The S.P. of the table = $(4m+70) = (200+70) = \text{Rs. } 270$

The C.P. of the chair = $3m = (3 \times 50) = \text{Rs. } 150$

The M.P. of the chair = $(3m+90) = (150+90) = \text{Rs. } 240$

The S.P. of the chair = $3.60m = (3.60 \times 50) = \text{Rs. } 180$

Shop Q

Let the M.P. of the table be Rs. $9n$. (*Note: $11.11\% = 1/9$*)

The C.P. of the table = Rs. 400

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The M.P. of the table = Rs. $9n$

The S.P. of the table = $\{(9n/9) \times 8\} = \text{Rs. } 8n$

The profit earned on selling the table = Rs. $(8n-400)$

The S.P. of the chair = Rs. $(8n-280)$

The C.P. of the chair = $\{(8n-280)-60\} = \text{Rs. } (8n-340)$

The M.P. of the chair = Rs. 420

The discount allowed on selling the chair = $\{420-(8n-280)\} = \text{Rs. } (700-8n)$

Now,

$$(8n-400)/(700-8n) = 4/1$$

$$8n - 400 = 2800 - 32n$$

$$40n = 3200$$

$$n = 80$$

Therefore,

The C.P. of the table = 400

The M.P. of the table = $9n = (9 \times 80) = 720$

The S.P. of the table = $8n = (8 \times 80) = \text{Rs. } 640$

The S.P. of the chair = $(8n-280) = (640-280) = \text{Rs. } 360$

The C.P. of the chair = $(8n-340) = (640-340) = \text{Rs. } 300$

Shop R

Let the marked price of the table be Rs. $6k$ (*Note: $16.66\% = 1/6$*)

The S.P. of the table = $\{(6k/6) \times 5\} = \text{Rs. } 5k$

The C.P. of the table = Rs. $(5k-70)$

The M.P. of the chair = Rs. $(6k-20)$

The S.P. of the chair = $75\% \text{ of } (6k-20) = \text{Rs. } (4.50k-15)$

The C.P. of the chair = $\{(4.50k-15)-60\} = \text{Rs. } (4.50k-75)$

Now, the ratio of the cost price of table and chair = $7:6$ (*Note: $14.28\% = 1/7$*)

$$(5k-70)/(4.50k-75) = 7/6$$

$$30k - 420 = 31.50k - 525$$

$$1.50k = 105$$

$$k = 70$$

Therefore,

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The M.P. of the table = $6k = (6 \times 70) = \text{Rs. } 420$

The S.P. of the table = $5k = (5 \times 70) = \text{Rs. } 350$

The C.P. of the table = $(5k - 70) = (350 - 70) = \text{Rs. } 280$

The M.P. of the chair = $(6k - 20) = (420 - 20) = \text{Rs. } 400$

The S.P. of the chair = $(4.50k - 15) = (4.50 \times 70 - 15) = (315 - 15) = \text{Rs. } 300$

The C.P. of the chair = $(4.50k - 75) = (315 - 75) = \text{Rs. } 240$

Shop P

Furniture	The cost price (in Rs.)	The marked price (in Rs.)	The selling price (in Rs.)
Table	200	300	270
Chair	150	240	180
Total	350	540	450

Shop Q

Furniture	The cost price (in Rs.)	The marked price (in Rs.)	The selling price (in Rs.)
Table	400	720	640
Chair	300	420	360
Total	700	1140	1000

Shop R

Furniture	The cost price (in Rs.)	The marked price (in Rs.)	The selling price (in Rs.)
Table	280	420	350
Chair	240	400	300
Total	520	820	650

Answer: B

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Shop R (after interchange)

Table

The C.P. = Rs. 240

The M.P. = Rs. 420

The S.P. = Rs. 350

Chair

The C.P. = Rs. 280

The M.P. = Rs. 400

The S.P. = Rs. 300

I: The profit earned on selling the chair = $(300-280) = \text{Rs. } 20$

So, I is definitely not true.

II: The mark-up percentage of the table = $\left[\frac{(420-240)}{240} \times 100\right]\% = \left\{\frac{(180)}{240} \times 100\right\}\% = 75\%$

So, II is definitely true.

III: The overall C.P. of table and chair together = $(240+280) = \text{Rs. } 520$

The overall S.P. of table and chair together = $(350+300) = \text{Rs. } 650$

The overall profit earned = $\left[\frac{(650-520)}{520} \times 100\right]\% = \left\{\frac{(130)}{520} \times 100\right\}\% = 25\%$

So, III is definitely true.

Therefore, both II and III are definitely true.

Hence, the correct answer is option B.

10. Questions

Shop P

Let the C.P. of table and chair be Rs. $4m$ and Rs. $3m$ respectively.

The C.P. of the table = Rs. $4m$

The M.P. of the table = 150% of $4m = \text{Rs. } 6m$

The S.P. of the table = Rs. $(4m+70)$

The discount allowed on the table = $\{6m-(4m+70)\} = \text{Rs. } (2m-70)$

The C.P. of the chair = Rs. $3m$

The M.P. of the chair = Rs. $(3m+90)$

The S.P. of the chair = $\frac{3}{5}$ of $6m = \text{Rs. } 3.60m$

The discount allowed on the chair = $\{(3m+90)-3.60m\} = \text{Rs. } (90-0.60m)$

Now,

$$(2m-70)/(90-0.60m) = 1/2$$

$$4m - 140 = 90 - 0.60m$$

$$4.60m = 230$$

$$m = 50$$

Therefore,

The C.P. of the table = $4m = (4 \times 50) = \text{Rs. } 200$

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The M.P. of the table = $6m = (6 \times 50) = \text{Rs. } 300$

The S.P. of the table = $(4m + 70) = (200 + 70) = \text{Rs. } 270$

The C.P. of the chair = $3m = (3 \times 50) = \text{Rs. } 150$

The M.P. of the chair = $(3m + 90) = (150 + 90) = \text{Rs. } 240$

The S.P. of the chair = $3.60m = (3.60 \times 50) = \text{Rs. } 180$

Shop Q

Let the M.P. of the table be Rs. $9n$. (*Note: $11.11\% = 1/9$*)

The C.P. of the table = Rs. 400

The M.P. of the table = Rs. $9n$

The S.P. of the table = $\{(9n/9) \times 8\} = \text{Rs. } 8n$

The profit earned on selling the table = Rs. $(8n - 400)$

The S.P. of the chair = Rs. $(8n - 280)$

The C.P. of the chair = $\{(8n - 280) - 60\} = \text{Rs. } (8n - 340)$

The M.P. of the chair = Rs. 420

The discount allowed on selling the chair = $\{420 - (8n - 280)\} = \text{Rs. } (700 - 8n)$

Now,

$$(8n - 400)/(700 - 8n) = 4/1$$

$$8n - 400 = 2800 - 32n$$

$$40n = 3200$$

$$n = 80$$

Therefore,

The C.P. of the table = 400

The M.P. of the table = $9n = (9 \times 80) = 720$

The S.P. of the table = $8n = (8 \times 80) = \text{Rs. } 640$

The S.P. of the chair = $(8n - 280) = (640 - 280) = \text{Rs. } 360$

The C.P. of the chair = $(8n - 340) = (640 - 340) = \text{Rs. } 300$

Shop R

Let the marked price of the table be Rs. $6k$ (*Note: $16.66\% = 1/6$*)

The S.P. of the table = $\{(6k/6) \times 5\} = \text{Rs. } 5k$

The C.P. of the table = Rs. $(5k - 70)$

The M.P. of the chair = Rs. $(6k - 20)$

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The S.P. of the chair = 75% of $(6k-20)$ = Rs. $(4.50k-15)$

The C.P. of the chair = $\{(4.50k-15)-60\}$ = Rs. $(4.50k-75)$

Now, the ratio of the cost price of table and chair = 7:6 (*Note: 14.28% = 1/7*)

$$(5k-70)/(4.50k-75) = 7/6$$

$$30k - 420 = 31.50k - 525$$

$$1.50k = 105$$

$$k = 70$$

Therefore,

The M.P. of the table = $6k = (6 \times 70)$ = Rs. 420

The S.P. of the table = $5k = (5 \times 70)$ = Rs. 350

The C.P. of the table = $(5k-70) = (350-70)$ = Rs. 280

The M.P. of the chair = $(6k-20) = (420-20)$ = Rs. 400

The S.P. of the chair = $(4.50k-15) = (4.50 \times 70 - 15) = (315-15)$ = Rs. 300

The C.P. of the chair = $(4.50k-75) = (315-75)$ = Rs. 240

Shop P

Furniture	The cost price (in Rs.)	The marked price (in Rs.)	The selling price (in Rs.)
Table	200	300	270
Chair	150	240	180
Total	350	540	450

Shop Q

Furniture	The cost price (in Rs.)	The marked price (in Rs.)	The selling price (in Rs.)
Table	400	720	640
Chair	300	420	360
Total	700	1140	1000

Shop R

Furniture	The cost price (in Rs.)	The marked price (in Rs.)	The selling price (in Rs.)
Table	280	420	350
Chair	240	400	300
Total	520	820	650

Answer: A

Shop Q (Table)

Let the increased C.P. be Rs. 11T (*Note: 9.09% = 1/11*)

So, the new S.P. = $\{(11T/11) \times 12\}$ = Rs. 12T

And, the new M.P. = Rs. (12T+20)

Now,

$$11T + 60 = 12T + 20$$

$$T = 40$$

So, the increased C.P. = 11T = (11×40) = Rs. 440

The initial C.P. of the chair = Rs. 400

So, the percentage increase in the initial C.P. = $[\{(440-400)/400\} \times 100]\% = \{(40/400) \times 100\}\% = 10\%$

Now,

$$2Z = 10$$

$$Z = 5$$

Hence, the correct answer is option A.

11. Questions

The ratio of the total number of people in village A and D = 75:100 = 3:4. Let it be 3m and 4m respectively.

Village A

The total number of people = 3m

The number of illiterate people = 40

The total number of literate people = (3m-40)

The number of post-graduate people = $\{(40/1) \times 3\}$ = 120

The number of graduate people = $\{(3m-40)-120\}$ = (3m-160)

Village D

The total number of people = 4m

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The number of illiterate people = 80

The total number of literate people = $(4m-80)$

The number of post-graduate people = $\{(80/2) \times 3\} = 120$

The number of graduate people = $\{(4m-80)-120\} = (4m-200)$

Now,

$$(4m-200) - (3m-160) = 80$$

$$4m - 280 = 3m - 160$$

$$m = 120$$

Now,

Let the number of literate people in village B and C be $2n$ and n respectively.

Village B

The total number of literate people = $2n$

The number of illiterate people = 200

The total number of people = $(2n+200)$

The number of graduate people = 200

The number of post-graduate people = $(2n-200)$

Village C

The total number of literate people = n

The number of illiterate people = 50

The total number of people = $(n+50)$

The number of post-graduate people = 120% of 50 = 60

The number of graduate people = $(n-60)$

Now,

$$(2n-200) - (n-60) = 10$$

$$2n - 210 = n - 60$$

$$n = 150$$

Village A

The total number of people = $3m = (3 \times 120) = 360$

The total number of literate people = $(3m-40) = (360-40) = 320$

The number of graduate people = $(3m-160) = (360-160) = 200$

Village B

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The total number of literate people = $2n = (2 \times 150) = 300$

The total number of people = $(2n+200) = (300+200) = 500$

The number of post-graduate people = $(2n-200) = (300-200) = 100$

Village C

The total number of literate people = $n = 150$

The total number of people = $(n+50) = (150+50) = 200$

The number of graduate people = $(n-60) = (150-60) = 90$

Village D

The total number of people = $4m = (4 \times 120) = 480$

The total number of literate people = $(4m-80) = (480-80) = 400$

The number of graduate people = $(4m-200) = (480-200) = 280$

Village E

The total number of people = 160

The number of illiterate people = 32

The total number of literate people = $(160-32) = 128$

The number of graduate people = $\{(128+16)/2\} = (144/2) = 72$

The number of post-graduate people = $\{(128-16)/2\} = (112/2) = 56$

Village	The total number of people	The number of literate people	The number of illiterate people	The number of graduate people	The number of post-graduate people
A	360	320	40	200	120
B	500	300	200	200	100
C	200	150	50	90	60
D	480	400	80	280	120
E	160	128	32	72	56
Total	1700	1298	402	842	456

Answer: C

Now, according to the question,

The total number of literate people in village A = 320

The total number of literate people in village B = 300

Using I

The number of male literates in village B = 160

So, the number of male literates in village A = $\{(160/80) \times 100\} = 200$

The number of female literates in village A = $(320-200) = 120$

The number of female literates in village B = $(300-160) = 140$

So, the required ratio = $120:140 = 6:7$

Therefore, I does satisfy.

Using II

The number of male literates in village B = 144

So, the number of male literates in village A = $\{(144/80) \times 100\} = 180$

The number of female literates in village A = $(320-180) = 140$

The number of female literates in village B = $(300-144) = 156$

So, the required ratio = $140:156 = 35:39$

Therefore, II does not satisfy.

Using III

The number of male literates in village B = 192

So, the number of male literates in village A = $\{(192/80) \times 100\} = 240$

The number of female literates in village A = $(320-240) = 80$

The number of female literates in village B = $(300-192) = 108$

So, the required ratio = $80:108 = 20:27$

Therefore, III does not satisfy.

Only I follow.

Hence, the correct answer is option C.

12. Questions

The ratio of the total number of people in village A and D = $75:100 = 3:4$. Let it be $3m$ and $4m$ respectively.

Village A

The total number of people = $3m$

The number of illiterate people = 40

The total number of literate people = $(3m-40)$

The number of post-graduate people = $\{(40/1) \times 3\} = 120$

The number of graduate people = $\{(3m-40)-120\} = (3m-160)$

Village D

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The total number of people = $4m$

The number of illiterate people = 80

The total number of literate people = $(4m-80)$

The number of post-graduate people = $\{(80/2) \times 3\} = 120$

The number of graduate people = $\{(4m-80)-120\} = (4m-200)$

Now,

$$(4m-200) - (3m-160) = 80$$

$$4m - 280 = 3m - 160$$

$$m = 120$$

Now,

Let the number of literate people in village B and C be $2n$ and n respectively.

Village B

The total number of literate people = $2n$

The number of illiterate people = 200

The total number of people = $(2n+200)$

The number of graduate people = 200

The number of post-graduate people = $(2n-200)$

Village C

The total number of literate people = n

The number of illiterate people = 50

The total number of people = $(n+50)$

The number of post-graduate people = 120% of 50 = 60

The number of graduate people = $(n-60)$

Now,

$$(2n-200) - (n-60) = 10$$

$$2n - 210 = n - 60$$

$$n = 150$$

Village A

The total number of people = $3m = (3 \times 120) = 360$

The total number of literate people = $(3m-40) = (360-40) = 320$

The number of graduate people = $(3m-160) = (360-160) = 200$

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Village B

The total number of literate people = $2n = (2 \times 150) = 300$

The total number of people = $(2n+200) = (300+200) = 500$

The number of post-graduate people = $(2n-200) = (300-200) = 100$

Village C

The total number of literate people = $n = 150$

The total number of people = $(n+50) = (150+50) = 200$

The number of graduate people = $(n-60) = (150-60) = 90$

Village D

The total number of people = $4m = (4 \times 120) = 480$

The total number of literate people = $(4m-80) = (480-80) = 400$

The number of graduate people = $(4m-200) = (480-200) = 280$

Village E

The total number of people = 160

The number of illiterate people = 32

The total number of literate people = $(160-32) = 128$

The number of graduate people = $\{(128+16)/2\} = (144/2) = 72$

The number of post-graduate people = $\{(128-16)/2\} = (112/2) = 56$

Village	The total number of people	The number of literate people	The number of illiterate people	The number of graduate people	The number of post-graduate people
A	360	320	40	200	120
B	500	300	200	200	100
C	200	150	50	90	60
D	480	400	80	280	120
E	160	128	32	72	56
Total	1700	1298	402	842	456

Answer: B

Village C

The total number of graduates = 90

The number of graduates who are employed = X

So, the number of graduates who are unemployed = $(90-X)$

The number of employed graduates who are employed in private sector = 50

The number of employed graduates who are employed in public sector = $\{(90-X) \times 3\} = (270-3X)$

Now,

$$50 + (270-3X) = X$$

$$4X = 320$$

$$X = 80$$

Therefore, the value of $\sqrt{(X+1)} = \sqrt{(80+1)} = \sqrt{81} = 9$.

Hence, the correct answer is option B.

13. Questions

The ratio of the total number of people in village A and D = 75:100 = 3:4. Let it be 3m and 4m respectively.

Village A

The total number of people = 3m

The number of illiterate people = 40

The total number of literate people = $(3m-40)$

The number of post-graduate people = $\{(40/1) \times 3\} = 120$

The number of graduate people = $\{(3m-40)-120\} = (3m-160)$

Village D

The total number of people = 4m

The number of illiterate people = 80

The total number of literate people = $(4m-80)$

The number of post-graduate people = $\{(80/2) \times 3\} = 120$

The number of graduate people = $\{(4m-80)-120\} = (4m-200)$

Now,

$$(4m-200) - (3m-160) = 80$$

$$4m - 280 = 3m - 160$$

$$m = 120$$

Now,

Let the number of literate people in village B and C be 2n and n respectively.

Village B

The total number of literate people = 2n

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The number of illiterate people = 200

The total number of people = $(2n+200)$

The number of graduate people = 200

The number of post-graduate people = $(2n-200)$

Village C

The total number of literate people = n

The number of illiterate people = 50

The total number of people = $(n+50)$

The number of post-graduate people = 120% of 50 = 60

The number of graduate people = $(n-60)$

Now,

$$(2n-200) - (n-60) = 10$$

$$2n - 210 = n - 60$$

$$n = 150$$

Village A

The total number of people = $3m = (3 \times 120) = 360$

The total number of literate people = $(3m-40) = (360-40) = 320$

The number of graduate people = $(3m-160) = (360-160) = 200$

Village B

The total number of literate people = $2n = (2 \times 150) = 300$

The total number of people = $(2n+200) = (300+200) = 500$

The number of post-graduate people = $(2n-200) = (300-200) = 100$

Village C

The total number of literate people = $n = 150$

The total number of people = $(n+50) = (150+50) = 200$

The number of graduate people = $(n-60) = (150-60) = 90$

Village D

The total number of people = $4m = (4 \times 120) = 480$

The total number of literate people = $(4m-80) = (480-80) = 400$

The number of graduate people = $(4m-200) = (480-200) = 280$

Village E

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The total number of people = 160

The number of illiterate people = 32

The total number of literate people = $(160-32) = 128$

The number of graduate people = $\{(128+16)/2\} = (144/2) = 72$

The number of post-graduate people = $\{(128-16)/2\} = (112/2) = 56$

Village	The total number of people	The number of literate people	The number of illiterate people	The number of graduate people	The number of post-graduate people
A	360	320	40	200	120
B	500	300	200	200	100
C	200	150	50	90	60
D	480	400	80	280	120
E	160	128	32	72	56
Total	1700	1298	402	842	456

Answer: A

Village F

The number of graduates = 120

The number of post-graduates = $(120-Y)$

So, the total number of literate people = $\{120+(120-Y)\} = (240-Y)$

The number of illiterate people = $\{(240-Y)/5 \times 1\} = (48-0.20Y)$

Now,

$$(240-Y) + (48-0.20Y) = 6Y$$

$$288 - 1.20Y = 6Y$$

$$7.20Y = 288$$

$$Y = 40$$

I: 40 is exactly divisible by both 4 and 10.

So, I is definitely true.

II: 40 is an even composite number.

So, II is definitely true.

III: The unit digit of 40 = 0

0 is not a prime number.

So, III is definitely not true.

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Therefore, both I and II is definitely true.

Hence, the correct answer is option A.

14. Questions

The ratio of the total number of people in village A and D = 75:100 = 3:4. Let it be 3m and 4m respectively.

Village A

The total number of people = 3m

The number of illiterate people = 40

The total number of literate people = (3m-40)

The number of post-graduate people = $\{(40/1) \times 3\} = 120$

The number of graduate people = $\{(3m-40)-120\} = (3m-160)$

Village D

The total number of people = 4m

The number of illiterate people = 80

The total number of literate people = (4m-80)

The number of post-graduate people = $\{(80/2) \times 3\} = 120$

The number of graduate people = $\{(4m-80)-120\} = (4m-200)$

Now,

$$(4m-200) - (3m-160) = 80$$

$$4m - 280 = 3m - 160$$

$$m = 120$$

Now,

Let the number of literate people in village B and C be 2n and n respectively.

Village B

The total number of literate people = 2n

The number of illiterate people = 200

The total number of people = (2n+200)

The number of graduate people = 200

The number of post-graduate people = (2n-200)

Village C

The total number of literate people = n

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The number of illiterate people = 50

The total number of people = $(n+50)$

The number of post-graduate people = 120% of 50 = 60

The number of graduate people = $(n-60)$

Now,

$$(2n-200) - (n-60) = 10$$

$$2n - 210 = n - 60$$

$$n = 150$$

Village A

The total number of people = $3m = (3 \times 120) = 360$

The total number of literate people = $(3m-40) = (360-40) = 320$

The number of graduate people = $(3m-160) = (360-160) = 200$

Village B

The total number of literate people = $2n = (2 \times 150) = 300$

The total number of people = $(2n+200) = (300+200) = 500$

The number of post-graduate people = $(2n-200) = (300-200) = 100$

Village C

The total number of literate people = $n = 150$

The total number of people = $(n+50) = (150+50) = 200$

The number of graduate people = $(n-60) = (150-60) = 90$

Village D

The total number of people = $4m = (4 \times 120) = 480$

The total number of literate people = $(4m-80) = (480-80) = 400$

The number of graduate people = $(4m-200) = (480-200) = 280$

Village E

The total number of people = 160

The number of illiterate people = 32

The total number of literate people = $(160-32) = 128$

The number of graduate people = $\{(128+16)/2\} = (144/2) = 72$

The number of post-graduate people = $\{(128-16)/2\} = (112/2) = 56$

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Village	The total number of people	The number of literate people	The number of illiterate people	The number of graduate people	The number of post-graduate people
A	360	320	40	200	120
B	500	300	200	200	100
C	200	150	50	90	60
D	480	400	80	280	120
E	160	128	32	72	56
Total	1700	1298	402	842	456

Answer: B

Village D

The total number of people = 480

The total number of people (males + females) who are above poverty line = $\frac{3}{4}$ of 480 = 360

The total number of people (males + females) who are below poverty line = $\frac{1}{4}$ of 480 = 120

Let the number of females who are above poverty line be k .

So, the number of males who are below poverty line = k

The number of males who are above poverty line = $(360-k)$

The number of females who are below poverty line = $(120-k)$

Now,

$$(360-k)/(120-k) = 7/1$$

$$360 - k = 840 - 7k$$

$$6k = 480$$

$$k = 80$$

So, the number of males who are above poverty line = $(360-k) = (360-80) = 280$

And, the number of females who are above poverty line = $k = 80$ @QuizByMickey

Therefore, the required difference = $(280-80) = 200$.

Hence, the correct answer is option B.

15. Questions

The ratio of the total number of people in village A and D = $75:100 = 3:4$. Let it be $3m$ and $4m$ respectively.

Village A

The total number of people = $3m$

The number of illiterate people = 40

The total number of literate people = $(3m-40)$

The number of post-graduate people = $\{(40/1) \times 3\} = 120$

The number of graduate people = $\{(3m-40)-120\} = (3m-160)$

Village D

The total number of people = $4m$

The number of illiterate people = 80

The total number of literate people = $(4m-80)$

The number of post-graduate people = $\{(80/2) \times 3\} = 120$

The number of graduate people = $\{(4m-80)-120\} = (4m-200)$

Now,

$$(4m-200) - (3m-160) = 80$$

$$4m - 280 = 3m - 160$$

$$m = 120$$

Now,

Let the number of literate people in village B and C be $2n$ and n respectively.

Village B

The total number of literate people = $2n$

The number of illiterate people = 200

The total number of people = $(2n+200)$

The number of graduate people = 200

The number of post-graduate people = $(2n-200)$

Village C

The total number of literate people = n

The number of illiterate people = 50

The total number of people = $(n+50)$

The number of post-graduate people = $120\% \text{ of } 50 = 60$

The number of graduate people = $(n-60)$

Now,

$$(2n-200) - (n-60) = 10$$

$$2n - 210 = n - 60$$

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$n = 150$

Village A

The total number of people = $3m = (3 \times 120) = 360$

The total number of literate people = $(3m - 40) = (360 - 40) = 320$

The number of graduate people = $(3m - 160) = (360 - 160) = 200$

Village B

The total number of literate people = $2n = (2 \times 150) = 300$

The total number of people = $(2n + 200) = (300 + 200) = 500$

The number of post-graduate people = $(2n - 200) = (300 - 200) = 100$

Village C

The total number of literate people = $n = 150$

The total number of people = $(n + 50) = (150 + 50) = 200$

The number of graduate people = $(n - 60) = (150 - 60) = 90$

Village D

The total number of people = $4m = (4 \times 120) = 480$

The total number of literate people = $(4m - 80) = (480 - 80) = 400$

The number of graduate people = $(4m - 200) = (480 - 200) = 280$

Village E

The total number of people = 160

The number of illiterate people = 32

The total number of literate people = $(160 - 32) = 128$

The number of graduate people = $\{(128 + 16) / 2\} = (144 / 2) = 72$

The number of post-graduate people = $\{(128 - 16) / 2\} = (112 / 2) = 56$

Village	The total number of people	The number of literate people	The number of illiterate people	The number of graduate people	The number of post-graduate people
A	360	320	40	200	120
B	500	300	200	200	100
C	200	150	50	90	60
D	480	400	80	280	120
E	160	128	32	72	56
Total	1700	1298	402	842	456

Answer: C

Village E

The total number of graduates = 72

Let the number of arts graduates be L.

So, the number of commerce graduates = $(L \times 3) = 3L$

And, the number of science graduates = $(3L + 16)$

Now,

$$L + 3L + (3L + 16) = 72$$

$$7L = 56$$

$$L = 8$$

Therefore, the number of science graduates = $(3L + 16) = (3 \times 8 + 16) = (24 + 16) = 40$

And, the number of arts graduates = $L = 8$

So, the required ratio = $40:8 = 5:1$.

Hence, the correct answer is option C.

16. Questions

Answer: D

Using statements I, II and III together

The initial quantity (milk + water) in vessel B = 180 ml

The initial quantity (milk + water) in vessel A = $\{(180/75) \times 100\} = 240$ ml

Let the initial quantity of milk in vessel A and B be $5x$ ml and $4x$ ml respectively.

So, the initial quantity of water in vessel A = $(240 - 5x)$ ml

And, the initial quantity of water in vessel B = $(180 - 4x)$ ml

Now,

$$(240 - 5x)/(180 - 4x) = 150/100$$

$$(240 - 5x)/(180 - 4x) = 3/2$$

$$480 - 10x = 540 - 12x$$

$$2x = 60$$

$$x = 30$$

So, the initial quantity of water in vessel A = $(240 - 5x) = (240 - 5 \times 30) = (240 - 150) = 90$ ml

And, the initial quantity of water in vessel B = $(180 - 4x) = (180 - 4 \times 30) = (180 - 120) = 60$ ml

The difference between the initial quantity of water in vessels A and B = $(90 - 60) = 30$ ml.

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Therefore, all statements I, II and III together are necessary.

Hence, the correct answer is option D.

17. Questions

Answer: D

The total work = 80 units

Using statements I, II and III together

The time taken by P and Q to complete the whole work together = $(80/9)$ days

The time taken by Q and R to complete the whole work together = $\{(8/70) \times 100\} = (80/7)$ days

Now, the efficiency of,

$$(P+Q) = \{80/(80/9)\} = 9$$

$$(Q+R) = \{80/(80/7)\} = 7$$

$$(P-R) = (9-7) = 2$$

Let the efficiency of P and R be $2k$ and k respectively.

$$2k - k = 2$$

$$k = 2$$

So, the efficiency of R = $k = 2$

Therefore, the time taken by R to complete the whole work alone = $(80/2) = 40$ days.

So, all statements I, II and III together are necessary.

Hence, the correct answer is option D.

18. Questions

Answer: C

The length of platform = 150 meters

The speed of train X = 90 m/sec

The speed of train Y = $\{(90/9) \times 8\} = 80$ m/sec (*Note: 12.50% = 1/8*)

Using statements I and II together

The length of the tunnel = 60 meters

Let the length of train X be L meters

Now, when train X crosses the tunnel,

$$(L+60)/90 = 4$$

$$L + 60 = 360$$

$$L = 300$$

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So, the length of train X = 300 meters

Therefore, the length of train Y = $\{(300/6) \times 5\} = 250$ meters

So, the time taken by train Y to cross the platform = $\{(250+150)/80\} = (400/80) = 5$ seconds.

Therefore, statement I and II together is sufficient.

Using statements I and III together

The speed of the man = 10 m/sec

Let the length of train X be K meters

Now, when train X crosses the man in opposite direction,

$$\{K/(90+10)\} = 3$$

$$K/100 = 3$$

$$K = 300$$

So, the length of train X = 300 meters

Therefore, the length of train Y = $\{(300/6) \times 5\} = 250$ meters

So, the time taken by train Y to cross the platform = $\{(250+150)/80\} = (400/80) = 5$ seconds.

Therefore, statement I and III together is sufficient.

So, either statement II or III together with statement I is sufficient.

Hence, the correct answer is option C.

19. Questions

Answer: D

Using statements I, II and III

The rate of interest offered in scheme N = 15% per annum

And, the simple interest received in scheme N = Rs. 22500

The time duration of scheme N = 3 years

Let the amount invested in scheme M and N be Rs. 4a and Rs. 5a respectively.

So, the simple interest received from scheme N = $(15 \times 3)\%$ of 5a = 45% of 5a = Rs. 2.25a

Now,

$$2.25a = 22500$$

$$a = 10000$$

So, the amount invested in scheme M = 4a = $(4 \times 10000) = \text{Rs. } 40000$

Now,

The rate of interest offered in scheme M = R% per annum

And, the compound interest received in scheme M = Rs. 17600

The time duration of scheme M = 2 years

So,

$$[(R+R)+\{(R^2/100)\}]\% = \{2R+(R^2/100)\}\% = \{(200R+R^2)/100\}\%$$

Now,

$$\{(200R+R^2)/100\} = \{(17600/40000)\times 100\}$$

$$200R + R^2 = (44 \times 100)$$

$$R^2 + 200R - 4400 = 0$$

$$R^2 + 220R - 20R - 4400 = 0$$

$$R(R+220) - 20(R+220) = 0$$

$$(R-20)(R+220) = 0$$

$$R = +20 \text{ or } -200$$

So, $R = 20$ (positive integer)

Therefore, all statements I, II and III together are necessary.

Hence, the correct answer is option D.

20. Questions

Answer: C

The speed of the river = 3 km/hr

Using statements I and II together

The ratio of the upstream speed of boat A and downstream speed of boat B = 80:100 = 4:5. Let it be $4v$ km/hr and $5v$ km/hr respectively.

So, the speed of boat A in still water = $(4v+3)$ km/hr

And, the speed of boat B in still water = $(5v-3)$ km/hr

Now, the downstream speed of boat A = $\{(4v+3)+3\} = (4v+6)$ km/hr

And, the upstream speed of boat B = $\{(5v-3)-3\} = (5v-6)$ km/hr

Now,

$$(5v-6)/(4v+6) = 1/2$$

$$10v - 12 = 4v + 6$$

$$6v = 18$$

$$v = 3$$

So, the speed of boat A in still water = $(4v+3) = (4 \times 3+3) = (12+3) = 15$ km/hr

Therefore, both statement I and II together are necessary.

Using statements II and III together

The ratio of the upstream speed of boat B and downstream speed of boat A = 1:2. Let it be s km/hr and $2s$ km/hr respectively.

So, the speed of boat A in still water = $(2s-3)$ km/hr

And, the speed of boat B in still water = $(s+3)$ km/hr

Now, the upstream speed of boat A = $\{(2s-3)-3\} = (2s-6)$ km/hr

And, the downstream speed of boat B = $\{(s+3)+3\} = (s+6)$ km/hr

So,

$$\{(2s-6) \times 5\} = \{(s+6) \times 4\}$$

$$10s - 30 = 4s + 24$$

$$6s = 54$$

$$s = 9$$

So, the speed of boat A in still water = $(2s-3) = (2 \times 9-3) = (18-3) = 15$ km/hr

Therefore, both statement II and III together are necessary.

So, either statements I and II together or II and III together is sufficient.

Hence, the correct answer is option C.

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